

Sawyer Welden

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SKILLS

Languages C#, Python, C++, R, Julia, JavaScript/TypeScript, SQL **Platforms** .NET, Angular, React, Azure, GCP, Docker, Power BI

Data and AI PyTorch, scikit-learn, xgboost, RAG and LLM applications, pandas, R Shiny

EXPERIENCE

Noregon Systems | *Software Engineer* | Greensboro, NC

Apr 2024 - Present

- Owned design and delivery of an AI-assisted repair-time estimation feature across Angular, .NET, SQL, and multiple model providers; added configurable prompts, human review controls, resumable batch processing, authorization, and isolated tests.
- Modernized six production services to the current .NET platform, replacing legacy persistence, mapping, mediator, and API-documentation infrastructure while preserving authentication, validation, serialization, and public contracts.
- Build full-stack systems supporting data from 300,000+ vehicles; reduced page-load time by 40%+ through SQL and Angular optimization and delivered customer-facing billing workflows across UI, API, and database layers.
- Authored 80+ pull requests across 17 repositories and completed 140+ code reviews across 20 repositories in one year, contributing to applications, services, databases, middleware, pipelines, tests, and documentation.

Wake Forest University School of Medicine | *Programmer 3* | Winston-Salem, NC

Jul 2021 - Dec 2023

- Built isolation-forest anomaly detection for the NIH-funded MoTrPAC clinical trial, replaced a legacy SAS pipeline with R, and created Shiny dashboards for clinicians investigating multivariate outliers across study sites.
- Trained posture-classification models in Python for a smart blood-pressure cuff and presented model improvements to healthcare professionals and investors.
- Co-developed the aorsf R package and wrote C++ acceleration for Cox regression, improving the computational efficiency and interpretability of oblique random survival forests.

Rochester Institute of Technology | *Graduate Research Assistant* | Rochester, NY

Jul 2019 - Dec 2021

- Built Julia and React tools for fair-division research, combining preference-model simulation, theorem development, and behavioral studies on Amazon Mechanical Turk.

SELECTED PUBLICATIONS

Rigdon, J., Saldana, S., Welden, S., et al. "Flexible outlier detection in multicenter clinical trials." *Journal of Clinical and Translational Science*, 10, e104 (2026).

Jaeger, B. C., Welden, S., et al. "Accelerated and Interpretable Oblique Random Survival Forests." *Journal of Computational and Graphical Statistics*, 33(1), 192-207 (2024).

Jaeger, B. C., Welden, S., Lenoir, K., and Pajewski, N. M. "aorsf: An R package for supervised learning using the oblique random survival forest." *Journal of Open Source Software*, 7(77), 4705 (2022).

SELECTED PROJECT

Pauper Data and Tools | *Competitive Magic: The Gathering* | sawyerwelden.com/pauper

- Built Python and JavaScript pipelines and interactive analyses for 1,000+ tournament decklists, including archetype normalization, metagame tracking, survivorship, card usage, pricing, and simulation tools.

EDUCATION

Rochester Institute of Technology MS Computer Science, 2021 **New College of Florida** BA Computer Science, 2019